

Combinations

Probability & Counting Worksheet · Grade 9–12

Name: _____

Date: _____

Learning Objectives

- Understand the difference between combinations and permutations, recognizing that order does not matter in combinations
- Apply the combination formula $C(n, r) = n! / [(n - r)! \cdot r!]$ using direct substitution
- Solve real-world word problems involving combinations by identifying n and r correctly

Problems

1. Evaluate the combination of 5 taken 2 at a time.

$$C(5, 2)$$

2. Evaluate the combination of 6 taken 3 at a time.

$$C(6, 3)$$

3. Evaluate the combination of 8 taken 5 at a time.

$$C(8, 5)$$

4. A teacher wants to select 2 students from a class of 7 to represent the school at a science fair. How many possible combinations of students can be chosen?

$$C(7, 2)$$

5. From a group of 5 volunteers — Gabby, Carly, Ivana, Alfredo, and Angelo — only 3 are chosen to join the Math Olympics. How many possible combinations can be formed?

$$C(5, 3)$$

6. A pizza shop offers 9 toppings. A customer wants to choose any 4 toppings for their pizza. How many



$$C(9, 4)$$

7. A basketball coach needs to choose 5 players from a roster of 12 to start a game. How many different starting lineups are possible?

$$C(12, 5)$$

8. A committee of 4 people is to be formed from a group of 10 candidates. In how many ways can the committee be selected? Then, determine how many more ways there would be if the committee size were increased to 5 people instead.

$$C(10, 4) \text{ and } C(10, 5)$$

9. A deck of 52 cards is used in a card game. A hand of 5 cards is dealt to a player. How many different 5-card hands are possible from a standard 52-card deck?

$$C(52, 5)$$

10. A school club has 6 boys and 5 girls. A committee of 4 members must be selected such that exactly 2 are boys and exactly 2 are girls. In how many ways can this committee be formed?

$$C(6, 2) \times C(5, 2)$$



Combinations — Answer Key

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Answer Key

1. Answer: 10

- Write the formula: $C(n, r) = n! / [(n - r)! \cdot r!]$
- Substitute $n = 5, r = 2$: $C(5, 2) = 5! / [(5 - 2)! \cdot 2!]$
- Simplify: $5! / (3! \cdot 2!) = (5 \times 4) / (2 \times 1) = 20 / 2 = 10$

2. Answer: 20

- Write the formula: $C(n, r) = n! / [(n - r)! \cdot r!]$
- Substitute $n = 6, r = 3$: $C(6, 3) = 6! / [(6 - 3)! \cdot 3!]$
- Simplify: $6! / (3! \cdot 3!) = (6 \times 5 \times 4) / (3 \times 2 \times 1) = 120 / 6 = 20$

3. Answer: 56

- Write the formula: $C(n, r) = n! / [(n - r)! \cdot r!]$
- Substitute $n = 8, r = 5$: $C(8, 5) = 8! / [(8 - 5)! \cdot 5!]$
- Simplify: $8! / (3! \cdot 5!) = (8 \times 7 \times 6) / (3 \times 2 \times 1) = 336 / 6 = 56$

4. Answer: 21

- Identify $n = 7$ (total students) and $r = 2$ (students to choose)
- Apply the formula: $C(7, 2) = 7! / [(7 - 2)! \cdot 2!]$
- Simplify: $(7 \times 6) / (2 \times 1) = 42 / 2 = 21$ possible combinations

5. Answer: 10

- Identify $n = 5$ (total volunteers) and $r = 3$ (students to choose)
- Apply the formula: $C(5, 3) = 5! / [(5 - 3)! \cdot 3!]$
- Simplify: $(5 \times 4 \times 3) / (3 \times 2 \times 1) = 60 / 6 = 10$ combinations

6. Answer: 126

- Identify $n = 9$ (total toppings) and $r = 4$ (toppings chosen)
- Apply the formula: $C(9, 4) = 9! / [(9 - 4)! \cdot 4!]$
- Expand numerator: $9 \times 8 \times 7 \times 6 = 3024$; denominator: $4 \times 3 \times 2 \times 1 = 24$
- $C(9, 4) = 3024 / 24 = 126$ combinations

7. Answer: 792

- Identify $n = 12$ (total players) and $r = 5$ (starters needed)
- Apply the formula: $C(12, 5) = 12! / [(12 - 5)! \cdot 5!]$
- Expand: $(12 \times 11 \times 10 \times 9 \times 8) / (5 \times 4 \times 3 \times 2 \times 1) = 95040 / 120 = 792$

8. Answer: $C(10,4) = 210$; $C(10,5) = 252$; difference = 42

- Calculate $C(10, 4)$: $10! / (6! \cdot 4!) = (10 \times 9 \times 8 \times 7) / (4 \times 3 \times 2 \times 1) = 5040 / 24 = 210$
- Calculate $C(10, 5)$: $10! / (5! \cdot 5!) = (10 \times 9 \times 8 \times 7 \times 6) / (5 \times 4 \times 3 \times 2 \times 1) = 30240 / 120 = 252$





- Difference: $252 - 210 = 42$ more ways with a committee of 5

9. Answer: 2,598,960

- Identify $n = 52$ and $r = 5$
- Apply the formula: $C(52, 5) = 52! / [(52 - 5)! \cdot 5!]$
- Expand numerator: $52 \times 51 \times 50 \times 49 \times 48 = 311,875,200$
- Denominator: $5! = 120$
- $C(52, 5) = 311,875,200 / 120 = 2,598,960$

10. Answer: 150

- Choose 2 boys from 6: $C(6, 2) = 6! / (4! \cdot 2!) = (6 \times 5) / (2 \times 1) = 15$
 - Choose 2 girls from 5: $C(5, 2) = 5! / (3! \cdot 2!) = (5 \times 4) / (2 \times 1) = 10$
 - Multiply the two combinations: $15 \times 10 = 150$ possible committees
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